

Interaction Region Characteristics for Midair Barehand Targeting on a Television

Jay Henderson
Carleton University
Ottawa, ON, Canada
jay.henderson@carleton.ca

Ali Neshati
University of Waterloo
Waterloo, ON, Canada
aneshati@uwaterloo.ca

Sachi Mizobuchi
Huawei Technologies
Markham, ON, Canada
sachi.mizobuchi@huawei.com

Wei Zhou
Huawei Technologies
Markham, ON, Canada
wei.zhou1@huawei.com

Daniel Vogel
University of Waterloo
Waterloo, ON, Canada
dvogel@uwaterloo.ca

Edward Lank
University of Waterloo
Waterloo, ON, Canada
lank@uwaterloo.ca

ABSTRACT

Recent computer vision advances can enable television control using midair barehand input, for which selecting targets using hand positions is a primary task. Yet, the size and position of a preferred 2D interaction region has not been specifically investigated for this setting. We report on a field study with people in front of their own television in their own home. Controlled variations of target position stimuli were presented while a camera records the natural hand position used by the participant in response. Based on hand and face landmarks, density plots define a preferred input region location and size using a human-scaled unit of face-widths. Distribution of hand positions relative to target stimuli reveals consistency and precision, suggesting an ability for users to map their input space to display space. We elicit an ideal input region from participants and provide a range of dimensions for designers implementing barehand input techniques.

CCS CONCEPTS

• **Human-centered computing** → *Pointing*.

KEYWORDS

Barehand input, Targeting, Smart TV

ACM Reference Format:

Jay Henderson, Ali Neshati, Sachi Mizobuchi, Wei Zhou, Daniel Vogel, and Edward Lank. 2023. Interaction Region Characteristics for Midair Barehand Targeting on a Television. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3585877>

1 INTRODUCTION

An emerging trend on Smart TVs is the ability to control the display using barehand gestures at a distance (see, for example, Huawei's X65, Hisense U7, and Samsung F-series displays). Users hold their

hand up in front of their body and the Smart TV, using optical or infrared cameras, tracks their hand position to recognize gestures and movement. A fundamental task that these smart displays require is point-and-click input or classical selection. Whether manipulating a video-on-demand interface, selecting an application, browsing the web, or performing a host of other tasks on Smart TVs, the classical selection operation represents a fundamental style of input applicable across a wide range of scenarios.

While there is related literature surrounding barehand and free-space input for external displays [9, 10, 13, 16, 22], there is still a lack of exploration on how users utilize around-body space, as the interaction area, to interact with the display. For instance, there is currently no firm consensus on how users segment the around-body interaction space to perform target selection tasks across a display. Are users consistent with their segmented space throughout interaction? In other words, if the user needs to select a target on the top left corner of the display, does the user consistently interact with a certain location relative to their body? The answers to these two questions can help us to define a space, relative to the user's body, as the interaction area for bare-hand input where users can comfortably and consistently perform target selection tasks.

Many prior works require extensive, state-of-the-art spatial tracking, such as a Vicon system, presented in extremely unrealistic and controlled laboratory settings. Though these systems undoubtedly represent highly accurate tracking, for the average user, such a system is impractical for usage in their home environment – and in practice, not how these input techniques are implemented in smart displays [1]. In particular, since camera capture is based on recognition within a 2D plane as opposed to a 3D raycast, there are a number of questions when designing for barehand pointing for smart display input, including:

1. Where should user input be located relative user's body and how large should the input range be so the user can comfortably interact with the display?
2. Does target arrangement on the display affect the size of the comfortable interaction area?
3. Will users consistently segment the interaction space based on target positions given no feedback?

Our study aims to create a user-defined natural input range for pointing at remote displays, by asking users to perform a series of prompted target selection tasks. This study was conducted in-situ in participants' homes using an RGB camera, similar to those that

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '23, April 23–28, 2023, Hamburg, Germany

© 2023 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3585877>

exist in commercial Smart TV displays [1], instead of ray-tracing technologies used in previous work [13, 22].

As a result, we elicit an ideal input region from participants and provide a range of dimensions for designers implementing barehand input – which, from the camera’s perspective, is on the order of two face widths wide by one face width high. Using face measurements allows us to calibrate for different distances from the display using face detection, a mature technology found on many modern digital cameras and smartphones.

2 RELATED WORK

This section will present a variety of past literature that primarily focus on understanding the characteristics of mapping in-air and barehand pointing techniques to output on a distant display.

2.1 Pointing Mechanisms

Though a fundamental requirement for most interactive systems, acquiring targets can be a challenge for smart displays since interaction is at a distance and, thus, more ambiguous than direct touch. Many prior works have extended and modified desktop input paradigms to midair input, allowing users to point and navigate a 2D GUI from a distance – via some form of hand-held device (such as a laser pointer or Wiimote) [3, 14, 15, 17, 20, 23], with an arm or wrist mounted device [5, 8], or by capturing the users barehand movements with external sensors [13, 19, 22, 24, 25] (e.g. Vicon, Kinect, other optical cameras).

2.1.1 Barehand Input. In terms of barehand input, traditional pointing consists of extending the index finger to localize the users intended target on a display, as used in Vogel and Balakrishnan’s work [22]. However, Brown et al. [2] suggest that the natural curvature of human fingers is unsuitable for raycasting to an external display. The barehand targeting research space has explored a variety of hand poses and gestures, for delimiting clicks (e.g. such as pinching or tapping a thumb/finger) [19, 22], and for UI manipulations through multiple finger raycasts [13].

2.2 Raycasting vs. Planar Segmentation

For much of the distal pointing literature, pointing is conducted by either raycasting or planar segmentation. The metaphor of raycasting depends on angular pointing data of an input mechanism [9] captured either using rotation of an using arm-worn or hand-held controller [5, 8, 17, 20] or via drawing a “ray” between two or more points [9, 10, 13, 15, 22]. Whereas planar segmentation involves positioning on an imaginary 2D plane in front of the user, followed by mapping the positional input to an external display [4, 7, 25]. Jota et al. [7] compared variants of absolutely mapped pointing to a distant display with a handheld controller, and found that for vertical single-dimensional targeting, rotational techniques exhibited better performance, but for tracing (steering) tasks, it was optimal to segment a plane in front of the user for input. Cockburn et al. also compared the two for relative mapping – and, similar to Jota et al.’s tracing task, found higher accuracy with spatial translation [4]. These works indicate promise for planar segmentation using RGB images that lack depth data for capturing angular raycasting information.

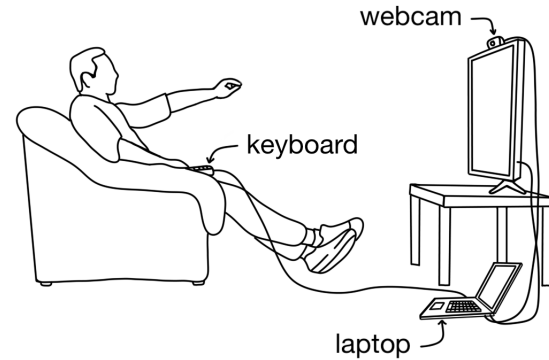


Figure 1: The experimental setup.

2.3 Defining the Input Region

Despite the extent of literature on mapping midair pointing, very few related works actually reveal the intended or measured size and location of their developed input region, with a few exceptions. For cursor mapping via a 2D plane, Cockburn et al. [4] defined an input region of 91 x 91 cm and when mapping to a midair keyboard, and Markussen et al. [11] defined an input region of 20 x 5.5 cm. Cockburn et al.’s participants noted that larger scale 2D movements helped them remember interaction locations, but that smaller scale movements were preferred due to less physical demand [4]. In the same vein, Hincapié-Ramos et al. compared two plane sizes for midair interactions, 35 x 35 cm and 25 x 25 cm, and found that the best plane for consumed endurance was the smaller of the two [6]. However, the authors found that the larger plane led to better performance – which, in part they suggest, could be due to tracking errors in distinguishing gestures.

In terms of location of input, for right handed users, Hincapié-Ramos et al. propose aligning the plane horizontally to the right side of the shoulder joint and to the vertical center halfway between the user’s shoulder and waist [6]. In addition, Poláček et al. found that users preferred interacting with their dominant hand and not in front of their face as it occluded their view [18]. More recently, Veras et al. proposed a spherical region centered on the shoulder of the users dominant side to allow for elbow anchoring. While a spherical region is suited to raycasting, it would likely pose challenges when detecting via RGB camera input.

3 EXPERIMENTAL PROTOCOL

In the introduction, the first question we posed involved input region. Ideally, we would like to understand where people naturally interact when they point at targets on large distant displays and what the natural size of this input region is. This, in turn, can guide the design of camera-based tracking algorithms. The goal of our experiment was, therefore, to define an appropriate input region for conducting barehand midair pointing on a distant display. To accomplish this, we conducted an in-situ observational study in participants’ homes to better understand how users perceive and acquire targets on a Smart TV or other in-home smart display.

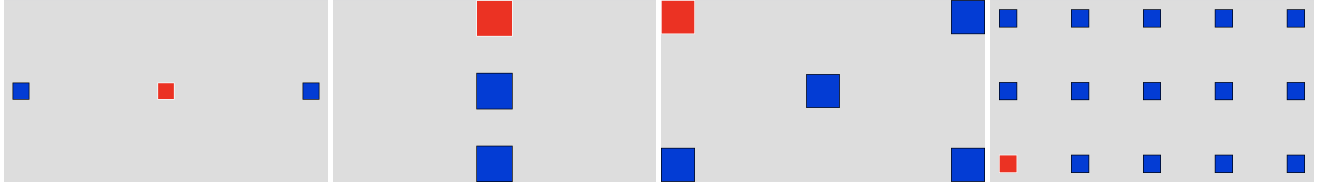


Figure 2: Example target task scenarios in our experiment, from left to right: B, D, G, I.

3.1 Participants

Twelve paid participants were recruited for the experiment, mean age was 34.83 (SD = 13.68, range [26, 59]). Five participants identified as women and the remaining seven identified as men. Eight participants were right handed, three were left handed, and one was ambidextrous but participated using their right hand. Our participants watched from 3 to 45 hours of TV per week, with a mean of 18.67 (SD = 12.30). Mean wingspan of participants was 166.48 (SD = 12.13). TV widths ranged from 95.3cm to 146.1cm, with a mean of 126.2cm (SD = 17.8cm). Distance from the participants' seats to the TV ranged from 136.0cm to 355.6cm, with a mean of 250.2cm (SD = 68.5cm). Lastly, the mean distance from the floor to the top of the TV was 159.4cm (SD = 39.6cm) with a range of 102cm to 227.3cm. Each participant signed an informed consent document form before starting the experiment.

3.2 Apparatus

Participants were required to have a laptop and TV, both with HDMI ports, to complete the study. They were provided with a Logitech C920S HD Pro Webcam, an HDMI cable, and a keyboard with a USB-A extension cord for providing input. Experiment setup was completed by participants through instructions from the researcher during a video call. Prior to commencing the study, the researcher verified their experimental setup via the video call. As an in-situ study, the experiment took place in each participant's own living accommodation, in the room where they watch and interact with a TV most often. The webcam was fastened to the top center of the TV, and connected to the participant's laptop, placed near the TV. The laptop was connected to the TV via HDMI, to the webcam via USB-A, and to the keyboard via USB-A. The setup is depicted in Figure 1. The experiment was presented through a web application written in Javascript, and hosted on ngrok.

3.3 Task and Stimuli

As in Figure 1, participants were instructed to sit in their typical manner while watching TV, to place the keyboard comfortably under their non-preferred hand, and complete the task as naturally and comfortably as possible with their preferred hand. Since the goal of the study was to determine *where* participants felt they should position their hand to acquire a target, continuous hand tracking was neither relevant nor required.

In a study on timeline manipulation, Zhang et al. [25] observed that the most natural target selection action for participants was a pinch on the desired target, i.e. open-hand in camera is a tracking state and closed fingers a select. We leveraged this pinch gesture: to simulate acquiring a target, participants were instructed to use

their index finger and thumb to select a target via a pinch gesture. Once they were in the position they felt should have acquired the target via pinch, participants pressed the space bar on the keyboard to indicate task completion. The application captured a picture of the participant's current positioning for processing, and the next trial was displayed.

Participants could lower their hand to rest, they could pause for as long as they wanted, and, in general, nothing was asked of the participant beyond holding and indicating selection via pinch and keyboard action.

Scenario	Alignment	# targets	# targets to select	Target dimensions
A	Horizontal	3	3	20% W x 20% H
B	Horizontal	3	3	10% W x 10% H
C	Horizontal	5	3	10% W x 10% H
D	Vertical	3	3	20% W x 20% H
E	Vertical	3	3	10% W x 10% H
F	Vertical	5	3	10% W x 10% H
G	Corners	5	5	20% W x 20% H
H	Corners	5	5	10% W x 10% H
I	Corners	15	5	10% W x 10% H

Table 1: Description of target task scenarios used in our experiment. W = display width, H = display height.

The targeting task mimicked typical target acquisition tasks that might be required during Smart TV interaction, e.g. on Netflix, with rectangular targets aligned in a grid-like fashion on the screen. To get an idea of the extremes of the elicited region, we designed 9 target scenarios comprised of three alignments *Horizontal*, *Vertical*, and *Corners*, with two different target sizes/densities (see Table 1). As most TVs expand target widths to fit the screen, target dimensions and locations were defined as percentages of the screen. These are presented in Table 1, with examples in Figure 2. Orderings of target scenarios were partially counter-balanced across participants, using a 9x9 Latin Square for the first nine participants, and the remaining three saw the next three orderings of a fully counter-balanced design. Scenarios with five targets were selected four times each and scenarios with three targets were selected six times each. Ordering of which targets to select were presented using either a 5x5 Latin Square for 5 targets or 3x3 Latin Square for 3 targets. Participants completed three blocks of each scenario.

In total, we collected:

$$\begin{aligned}
 &12 \text{ participants} \times 3 \text{ blocks} \times \\
 &(6 \text{ tasks} \times 3 \text{ targets} \times 6 \text{ trials} + 3 \text{ tasks} \times 5 \text{ targets} \times 4 \text{ trials}) \\
 &= 6048 \text{ images}
 \end{aligned}$$

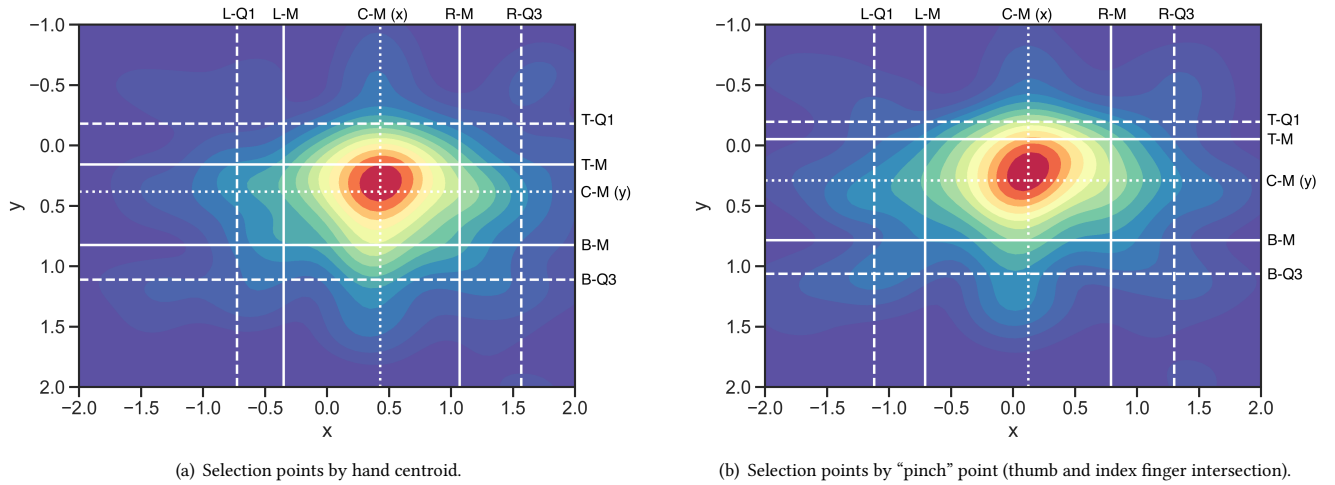


Figure 3: Density plots of all selection points. x- and y-axes are number of face widths from the annotated face center. Note: to aggregate left and right handed participants, the x-axis becomes increasingly positive to the user’s dominant, or interacting, side. Each line indicates a key position along each axis for dividing the input region, as discussed in Table 2.

Label	Type	Axis	Hand Center	Pinch	Targets
C-M (x)	Median	x	0.45	0.10	center
C-M (y)	Median	y	0.40	0.30	center
L-M	Median	x	-0.35	-0.70	top-left, left, bottom-left
R-M	Median	x	1.05	0.80	top-right, right, bottom-right
T-M	Median	y	0.15	-0.05	top-left, top, top-right
B-M	Median	y	0.80	0.80	bottom-left, bottom, bottom-right
L-Q1	Q1	x	-0.75	-1.10	top-left, left, bottom-left
R-Q3	Q3	x	1.55	1.30	top-right, right, bottom-right
T-Q1	Q1	y	-0.20	-0.20	top-left, top, top-right
B-Q3	Q3	y	1.10	1.05	bottom-left, bottom, bottom-right

Table 2: Key positions in Figure 3, used for defining an appropriate input region, rounded to the nearest 0.05 FW (face widths).

3.4 Results

In this results section, we present our analysis around three primary questions:

- Where should the input region be located?
- How large should the input region be?
- Can we learn anything about input region mapping by analyzing user interaction location?

To perform this analysis, images captured from participants were manually annotated and verified individually by a researcher and an external party for validation, through a customized JavaScript web application. We removed 883 images that did not properly depict selection point, either due to a click error on they keyboard or if the participant was in motion during image capture, causing a blurry image – leaving a total of 5165 images for our analysis. Face detection was completed automatically using Haar Cascades in OpenCV for JavaScript and then verified by either the researcher or external party. If the detected face was not properly recognized,

a manual correction was performed. For the majority of images, annotation comprised of selecting the top-left corner and bottom right corner of the user’s hand to obtain a height and width, and marked the “pinch location” (where the index finger and thumb meet). These landmarks were then correlated with the stimulus images, i.e. the image presented to the participant (see Figure 2).

Our analysis exclusively uses face widths (FW throughout the paper) as a distance metric for two primary reasons. First, our images were captured using a simple RGB camera. Without depth information, it is challenging to extract physical measurements of a user’s barehand position and elicited range of motion (e.g. in cm or inches). Face width provides a reliable distance metric that allows for aggregation of participant data. Since it is relative to the user’s position, it is robust to both depth and repositioning. We can assume the closer the user is to the camera, the larger their face width will be in pixels. Therefore, via the RGB output the user’s range of movement in pixels will be greater the closer one gets to the camera. Second, as opposed to using shoulder width or

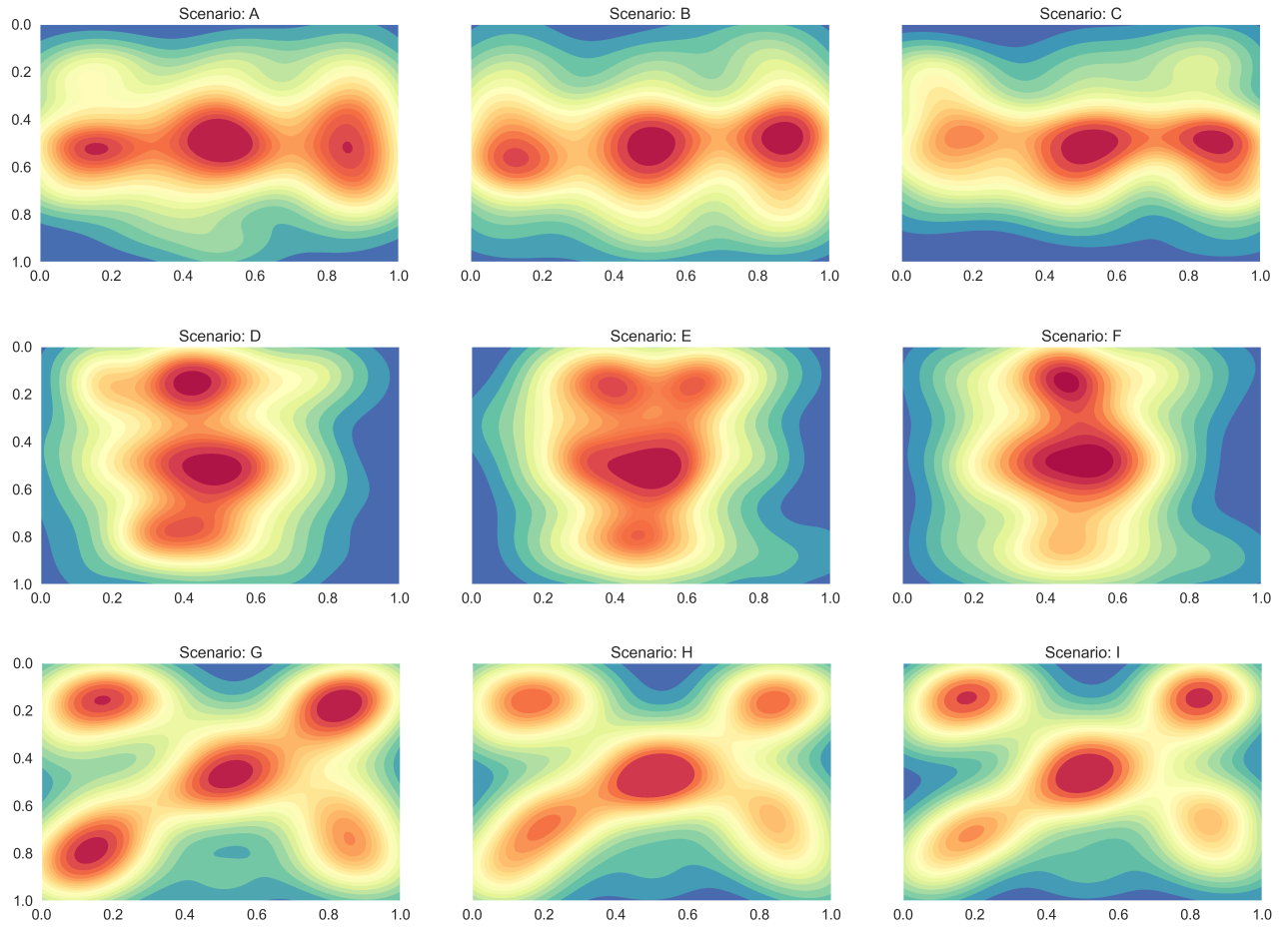


Figure 4: Density plots for target selection location from center of face, normalized on a scale of 0-1 per participant. The x-axis increases to the user’s dominant, or interacting, side.

hand width, face detection using an RGB image is a very mature technology. Thus, interaction designers and engineers can easily use such a metric when incorporating barehand midair input in their system, regardless of technical ability or available hardware.

3.4.1 Locating the Input Region. Leveraging coordinates relative to participants’ faces, we created a density plot of the centroids of all participant selection actions, shown in Figure 3. The origin of the coordinate system depicted is the center of the bounding box of the participant’s face, which is typically located at the middle of the nose. If we assume this represents the centroid of all input (there were an equal number of top-bottom/left-right/corner-corner inputs for each participant), the density plot provides an estimate of the perceived center of the pointing task.

We find that participants centered their input region just below the vertical face center, as can be seen in Figure 3. Leveraging the pinch point, this region tends to be near centered on the participant’s face/body (at 0.1 FW). The centroid of the hand (a landmark that was used in optical tracking in the next experiment) is located approximately 0.45 FW from center toward the dominant hand

side for participants. We also performed an analysis of left-handed and right-handed participants separately using the same scale but inverted. Results were consistent across handedness.

3.4.2 Sizing the Input Region. To size the input region, we present key positions in Table 2 and Figure 3, in order to develop an appropriate range input region sizes for interaction developers. The dotted line indicates the median center point when selecting “center” targets. The solid line depicts the median point when selecting a target that borders or is close to the outer edge of the display (e.g. the bottom line is the median for targets labelled “bottom-left”, “bottom”, and “bottom-right”). The dashed line indicates the interquartile (IQR) range for all targets selected.

Considering hand centroid, by subtracting the medians and interquartile ranges of selection points, the defined range horizontally is between 1.30 to 2.30 FW. Vertically, the range is smaller, between 0.65 to 1.30 FW. In terms of pinch position, the horizontal range is slightly larger, likely due to hand rotation, between 1.50 to 2.40 FW. However, the vertical range is slightly smaller at 0.6 to 1.25 FW.

3.4.3 Precision Through Density Plots. A final analysis we performed examines precision of participants in terms of their behaviour. Figure 4 presents density plots of all click locations for all users in the data set separated by scenario (top = horizontal, mid = vertical, bottom = corners).

3.5 Summary

Recall that the goal of our experiment was to elicit a natural input region from participants. Considering Figure 3 and Table 2, we identify a natural input region in terms of participant face widths (FW). When detection is through pinch location, participants naturally interact in a region just below the face center (0.1 FW offset) and when tracking the centroid, the region is offset slightly toward the dominant hand side of the face (0.45 FW offset). Perhaps one of the most interesting results from our study, was the relative precision of participants in terms of targeting location (see Figure 4). Since each of the scenarios' subfigures by scenario appear to have distinct sections of increased density per target, this suggests that participants have a natural ability to segment a 2D space in front of them.

4 DISCUSSION

In section 3, we present a user defined two-dimensional region for interaction with a smart display, such as a TV. To start, while prior work centers the midair input region around the edge of the users shoulder [6] or elbow [21], in our study, the median center of the user's pinch position lies very close to the horizontal face center (Figure 3(b) and 2). Therefore, our results indicate that users tend to imagine the input region aligned with the center of their body. We predict this is likely attributed to the increased strength toward the center of the body [6], which would ideally result in a less physically demanding interaction. That being said, if hand tracking is accomplished by tracking via hand centroid, as in our second experiment, we show that this region should be slightly offset to the users preferred hand's side (by 0.45 FW), due to the hand facing inward toward the body. On the vertical axis, we found users tend to center the input slightly above the bottom of their face, however, this should be used with caution as to not obstruct the user's view [18].

While few prior works provide planar dimensions for their input regions, they have ranged from width of 20cm to 91cm and height of 5.5cm to 91cm. In terms of the input region size, we provide a range – between 1.30 and 2.30 face widths wide and 0.65 to 1.30 FW tall when measured by hand centroid ($1.50 - 2.40 \times 0.60 - 1.25$ FW for pinch). Given the average white human face is approximately 13.3cm wide [12], this corresponds to a region approximately 17.3 - 30.3cm wide and 8.6 - 17.3cm tall ($20.0 - 32.0\text{cm} \times 8.0 - 16.6\text{cm}$ for pinch).

We present a range of widths, as prior work states, because in terms of physical demand, a smaller region would be better [6], however, the smaller the region, the more challenging it becomes for a system to distinguish input positions – particularly in more practical deployments that lack extensive hardware such as a Vi-con system. Thus, we encourage designers to adapt the region as necessary – but we recommend remaining within the region when possible.

5 CONCLUSION

To summarize our contributions, first, we present an in-situ study that reveals a user elicited 2D input region for targeting on a Smart TV. We open the space to future work to determine whether our defined region is transferable to additional poses (seated vs. standing) and into different contexts, such as in-vehicle input or for smart speaker devices. An additional next step would be to determine how best to leverage input region to map user input onto display space, i.e. an absolute versus relatively mapped region. Our results provide valuable insights for designing distal pointing interactions for smart displays, notably, an ideal range of dimensions for barehand input techniques.

REFERENCES

- [1] 2020. What is gesture control in Samsung F series smart TV?: Samsung Support. <https://www.samsung.com/in/support/tv-audio-video/what-is-gesture-control-in-samsung-f-series-smart-tv/>
- [2] Michelle A. Brown and Wolfgang Stuerzlinger. 2016. Exploring the Throughput Potential of In-Air Pointing. In *18th International Conference on Human-Computer Interaction. Interaction Platforms and Techniques, Part II (HCI International '16, Vol. LNCS 9732)*, 13–24. https://doi.org/10.1007/978-3-319-39516-6_2
- [3] Xiang Cao and Ravin Balakrishnan. 2004. VisionWand: Interaction Techniques for Large Displays Using a Passive Wand Tracked in 3D. *ACM Trans. Graph.* 23, 3 (aug 2004), 729. <https://doi.org/10.1145/1015706.1015788>
- [4] A. Cockburn, P. Quinn, C. Gutwin, G. Ramos, and J. Looser. 2011. Air Pointing: Design and Evaluation of Spatial Target Acquisition with and without Visual Feedback. *Int. J. Hum.-Comput. Stud.* 69, 6 (June 2011), 401–414. <https://doi.org/10.1016/j.ijhcs.2011.02.005>
- [5] Faizan Haque, Mathieu Nancel, and Daniel Vogel. 2015. *Myopoint: Pointing and Clicking Using Forearm Mounted Electromyography and Inertial Motion Sensors*. Association for Computing Machinery, New York, NY, USA, 3653–3656. <https://doi.org/10.1145/2702123.2702133>
- [6] Juan David Hincapié-Ramos, Xiang Guo, Paymahn Moghadasian, and Pourang Irani. 2014. Consumed Endurance: A Metric to Quantify Arm Fatigue of Mid-Air Interactions. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Toronto, Ontario, Canada) (CHI '14). Association for Computing Machinery, New York, NY, USA, 1063–1072. <https://doi.org/10.1145/2556288.2557130>
- [7] Ricardo Jota, Miguel A. Nacenta, Joaquim A. Jorge, Sheelagh Carpendale, and Saul Greenberg. 2010. A Comparison of Ray Pointing Techniques for Very Large Displays. In *Proceedings of Graphics Interface 2010* (Ottawa, Ontario, Canada) (GI '10). Canadian Information Processing Society, CAN, 269–276.
- [8] Keiko Katsuragawa, Krzysztof Pietroszek, James R. Wallace, and Edward Lank. 2016. Watchpoint: Freehand Pointing with a Smartwatch in a Ubiquitous Display Environment. In *Proceedings of the International Working Conference on Advanced Visual Interfaces* (Bari, Italy) (AVI '16). Association for Computing Machinery, New York, NY, USA, 128–135. <https://doi.org/10.1145/2909132.2909263>
- [9] Regis Kopper, Doug A. Bowman, Mara G. Silva, and Ryan P. McMahan. 2010. A human motor behavior model for distal pointing tasks. *International Journal of Human-Computer Studies* 68, 10 (2010), 603–615. <https://doi.org/10.1016/j.ijhcs.2010.05.001>
- [10] Regis Kopper, Mara G. Silva, Ryan P. McMahan, and D. Bowman. 2008. Increasing the Precision of Distant Pointing for Large High-Resolution Displays.
- [11] Anders Markussen, Mikkel Rønne Jakobsen, and Kasper Hornbæk. 2014. Vulture: A Mid-Air Word-Gesture Keyboard. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Toronto, Ontario, Canada) (CHI '14). Association for Computing Machinery, New York, NY, USA, 1073–1082. <https://doi.org/10.1145/2556288.2556964>
- [12] Zachary Raffensperger, Mary L. Marazita, Seth Weinberg. 2017. 3D Facial Norms Database. *FaceBase Consortium* (2017). <https://doi.org/10.25550/1WWP>
- [13] Fabrice Matulic and Daniel Vogel. 2018. *Multiray: Multi-Finger Raycasting for Large Displays*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3173574.3173819>
- [14] Mathieu Nancel, Olivier Chapuis, Emmanuel Pietriga, Xing-Dong Yang, Pourang P. Irani, and Michel Beaudouin-Lafon. 2013. High-Precision Pointing on Large Wall Displays Using Small Handheld Devices. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Paris, France) (CHI '13). Association for Computing Machinery, New York, NY, USA, 831–840. <https://doi.org/10.1145/2470654.2470773>
- [15] Mathieu Nancel, Emmanuel Pietriga, Olivier Chapuis, and Michel Beaudouin-Lafon. 2015. Mid-Air Pointing on Ultra-Walls. *ACM Trans. Comput.-Hum. Interact.* 22, 5, Article 21 (Aug. 2015), 62 pages. <https://doi.org/10.1145/2766448>

- [16] Mathieu Nancel, Julie Wagner, Emmanuel Pietriga, Olivier Chapuis, and Wendy Mackay. 2011. Mid-Air Pan-and-Zoom on Wall-Sized Displays. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vancouver, BC, Canada) (CHI '11). Association for Computing Machinery, New York, NY, USA, 177–186. <https://doi.org/10.1145/1978942.1978969>
- [17] Krzysztof Pietroszek, Anastasia Kuzminykh, James R. Wallace, and Edward Lank. 2014. Smartcasting: A Discount 3D Interaction Technique for Public Displays. In *Proceedings of the 26th Australian Computer-Human Interaction Conference on Designing Futures: The Future of Design* (Sydney, New South Wales, Australia) (OzCHI '14). Association for Computing Machinery, New York, NY, USA, 119–128. <https://doi.org/10.1145/2686612.2686629>
- [18] Ondrej Polacek, Martin Klima, Adam J. Sporka, Pavel Zak, Michal Hradis, Pavel Zemcik, and Vaclav Prochazka. 2012. A Comparative Study on Distant Free-Hand Pointing. In *Proceedings of the 10th European Conference on Interactive TV and Video* (Berlin, Germany) (EuroTV '12). Association for Computing Machinery, New York, NY, USA, 139–142. <https://doi.org/10.1145/2325616.2325644>
- [19] Matthias Schwaller and Denis Lalanne. 2013. Pointing in the Air: Measuring the Effect of Hand Selection Strategies on Performance and Effort. https://doi.org/10.1007/978-3-642-39062-3_53
- [20] Shaishav Siddhpuria, Sylvain Malacria, Mathieu Nancel, and Edward Lank. 2018. *Pointing at a Distance with Everyday Smart Devices*. Association for Computing Machinery, New York, NY, USA, 1–11. <https://doi.org/10.1145/3173574.3173747>
- [21] Rafael Veras, Gaganpreet Singh, Farzin Farhadi-Niaki, Ritesh Udhani, Parth Pradeep Patekar, Wei Zhou, Pourang Irani, and Wei Li. 2021. *Elbow-Anchored Interaction: Designing Restful Mid-Air Input*. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/3411764.3445546>
- [22] Daniel Vogel and Ravin Balakrishnan. 2005. Distant Freehand Pointing and Clicking on Very Large, High Resolution Displays. In *Proceedings of the 18th Annual ACM Symposium on User Interface Software and Technology* (Seattle, WA, USA) (UIST '05). Association for Computing Machinery, New York, NY, USA, 33–42. <https://doi.org/10.1145/1095034.1095041>
- [23] Chadwick A. Wingrave, Brian Williamson, Paul D. Varcholik, Jeremy Rose, Andrew Miller, Emiko Charbonneau, Jared Bott, and Joseph J. LaViola. 2010. The Wiimote and Beyond: Spatially Convenient Devices for 3D User Interfaces. *IEEE Computer Graphics and Applications* 30, 2 (2010), 71–85. <https://doi.org/10.1109/MCG.2009.109>
- [24] Ionuț-Alexandru Zaiți, Ștefan-Gheorghe Pentiuc, and Radu-Daniel Vatavu. 2015. On free-hand TV control: experimental results on user-elicited gestures with Leap Motion. *Personal and Ubiquitous Computing* 19, 5 (2015), 821–838. <https://doi.org/10.1007/s00779-015-0863-y>
- [25] Futian Zhang, Sachi Mizobuchi, Wei Zhou, Taslim Arefin Khan, Wei Li, and Edward Lank. 2021. Leveraging CD Gain for Precise Barehand Video Timeline Browsing On Smart Displays. In *Human-Computer Interaction – INTERACT 2021: 18th IFIP TC 13 International Conference, Bari, Italy, August 30 – September 3, 2021, Proceedings, Part IV* (Bari, Italy). Springer-Verlag, Berlin, Heidelberg, 72–91. https://doi.org/10.1007/978-3-030-85610-6_5